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About Me : I am a student with a keen interest in web application development, backend engineering, and database management. I enjoy building practical tools that simplify real-world processes, focusing on secure architecture and user-friendly design.

Description

This project is a dynamic Placement Portal web application designed to connect Students, Companies, and Administrators. It establishes an integrated platform handling role-based sign-on, recruitment drive creations, candidate applications, profile management, and administrative moderation actions.

AI/LLM models were used roughly 10-15% of the time, primarily for accelerating the UI design process, refining Jinja2 templating syntax, and resolving minor database relationship bugs.

AI/LLM usage

I used Google Gemini to assist with HTML templates, CSS styling, and debugging SQLAlchemy queries. The extent of AI usage is around 10-15%, mainly for code suggestions and minor optimizations. All core logic, integration, and final implementation were done manually.

Technologies used

- 1) **Python & Flask** : Backend development and request handling
- 2) **SQLite & SQLAlchemy** : Database management with ORM
- 3) **Flask-SQLAlchemy** : Integration between Flask and database
- 4) **Werkzeug** : Security features like password hashing
- 5) **Jinja2** : Dynamic frontend templating

DB Schema Design

Tables:

1. AdminUser — admin credentials (id, username, password_hash, is_active)
2. StudentUser — student details (id, full_name, email, phone_number, department, cgpa, account_status)
3. StudentProfile — resume and skills (id, student_id, resume_filename, skills, about_me)
4. CompanyAccount — company details (id, company_name, contact_email, approval_state, account_state)
5. PlacementDrive — job drives (id, company_id, drive_title, job_role, deadline, drive_status)
6. Application — job applications (id, student_id, drive_id, current_status, recruiter_remark)

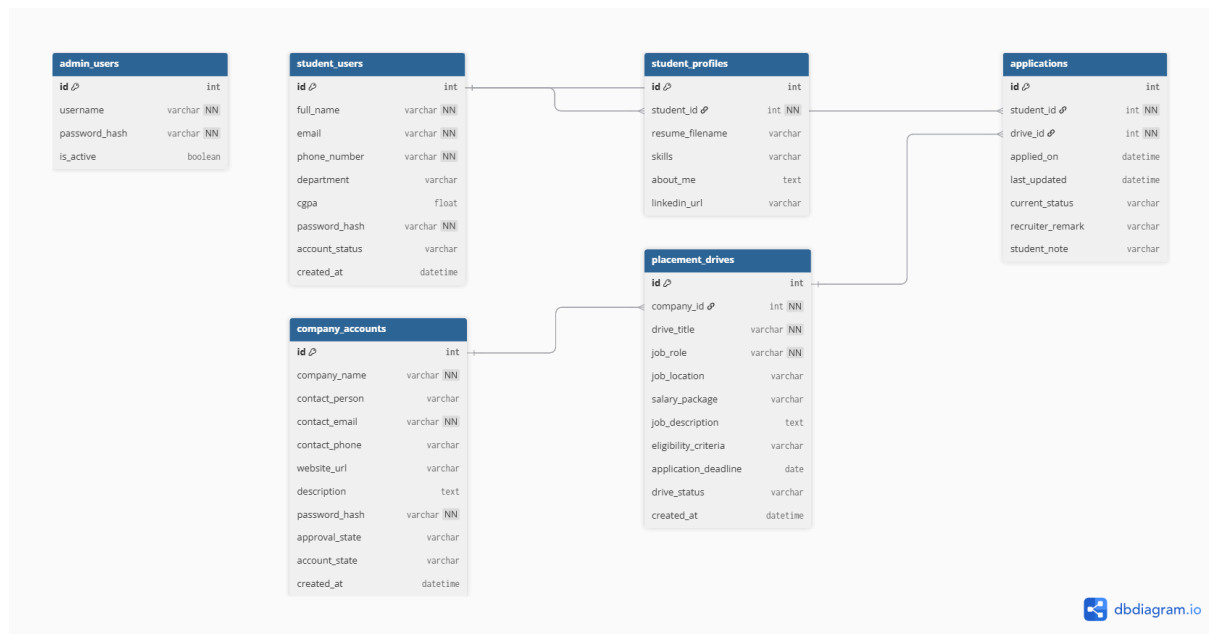
Relationships:

One-to-One → StudentUser → StudentProfile

One-to-Many → StudentUser → Application

One-to-Many → CompanyAccount → PlacementDrive

One-to-Many → PlacementDrive → Application



Architecture and Features

Architecture Overview:

- app.py – main Flask application entry point
- /models – database models using SQLAlchemy
- /templates – Jinja2 HTML templates
- /static – CSS and uploaded files

Implemented Features:

- Role-based user registration and login
- Placement drive creation and management
- Student profile and resume upload
- Application submission with deadline control
- Admin dashboard for approval and moderation

Additional Features:

- Dynamic application status updates (Applied, Shortlisted, Placed)
- Unique constraints to prevent duplicate applications

Video

Recorded Video Link:

https://drive.google.com/file/d/1XHFbBEzoE8uKMUf5n9lXcl4iKoo5g_VQ/view?usp=sharing